

Geoffrey J. Stanley

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CANADA

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EDUCATION

- Doctor of Philosophy, University of Oxford, UK (2018)
Atmospheric, Oceanic, & Planetary Physics
Thesis: *Tales from Topological Oceans*.
Supervisor: David Marshall
- Master of Science, University of Victoria, Canada (2013)
School of Earth and Ocean Sciences
Thesis: *From winds to eddies to diapycnal mixing of the deep ocean: the abyssal meridional overturning circulation driven by the surface wind-stress*.
Supervisors: Oleg Saenko and Andrew Weaver
- Bachelor of Mathematics, University of Waterloo, Canada (2011)
Applied Mathematics (Mathematical Physics) & Pure Mathematics Double Major
Dean's Honours List, With Distinction

ACADEMIC EMPLOYMENT

- Apr 2024 - present: Research Scientist
 - Canadian Centre for Climate Modelling and Analysis, Environment and Climate Change Canada
 - Lead for the Ocean Working Group
- Oct 2022 - Mar 2024: Banting Postdoctoral Researcher
 - School of Earth and Ocean Sciences, University of Victoria, Canada
 - Supervisors: Adam Monahan, James Anstey
- Jul 2021 - Sept 2022: Visiting Fellow
 - Research School of Earth Sciences, Australian National University, Australia
 - Supervisor: Andy Hogg
- Jan 2019 - Sept 2022: Postdoctoral Fellow
 - School of Mathematics and Statistics, University of New South Wales, Australia
 - Supervisor: Trevor McDougall

JOURNAL ARTICLES

- Drake, H. F., Bailey, S., Dussin, R., Griffies, S. M., Krasting, J., MacGilchrist, G., **Stanley, G.**, Tesdal, J., & Zika, J. D. (2025). Water Mass Transformation Budgets in Finite-Volume Generalized Vertical Coordinate Ocean Models. *Journal of Advances in Modeling Earth Systems*, 17(3), e2024MS004383. <https://doi.org/10.1029/2024MS004383>
- Bennetts, L. G., Shakespeare, C. J., Vreugdenhil, C. A., Foppert, A., Gayen, B., Meyer, A., Morrison, A. K., Padman, L., Phillips, H. E., Stevens, C. L., Toffoli, A., Constantinou, N. C., Cusack, J. M., Cyriac, A., Doddridge, E. W., England, M. H., Evans, D. G., Heil, P., Hogg, A. McC., ... **Stanley, G. J.** (2024). Closing the Loops on Southern Ocean Dynamics: From the Circumpolar Current to Ice Shelves and From Bottom Mixing to Surface Waves. *Reviews of Geophysics*, 62(3), e2022RG000781. <https://doi.org/10.1029/2022RG000781>

- Lang Y., **Stanley, G. J.**, & McDougall, T. J.. (2023). Spurious Dianeutral Advection and Methods for Its Minimization. *Journal of Physical Oceanography*. <https://doi.org/10.1175/JPO-D-22-0174.1>
- Bisits, J. I., **Stanley, G. J.**, & Zika, J. D. (2022). Can We Accurately Quantify a Lateral Diffusivity from A Single Tracer Release? *Journal of Physical Oceanography*. <https://doi.org/10.1175/JPO-D-22-0145.1>
- Malan, N., Roughan, M., **Stanley, G. J.**, Holmes, R., & Li, J. (2022). Quantifying cross-shelf transport in the East Australian Current System: A budget-based approach. *Journal of Physical Oceanography*, 52(10), 2555-2572. <https://doi.org/10.1175/JPO-D-21-0193.1>
- **Stanley, G. J.**, & Marshall, D. P. (2022). Why Mean Potential Vorticity Cannot Be Materially Conserved in the Eddying Southern Ocean. *Journal of Physical Oceanography*, 52(8), 1629-1654. <https://doi.org/10.1175/JPO-D-21-0195.1>
- **Stanley, G. J.**, McDougall, T. J., & Barker, P. M. (2021). Algorithmic improvements to finding approximately neutral surfaces. *Journal of Advances in Modeling Earth Systems*, 13(5), e2020MS002436. <https://doi.org/10.1029/2020MS002436>
- McDougall, T. J., Barker, P. M., & **Stanley, G. J.** (2021). Spice variables and their use in physical oceanography. *Journal of Geophysical Research: Oceans*, 126(2), e2019JC015936. <https://doi.org/10.1029/2019JC015936>
- Lang, Y., **Stanley, G. J.**, McDougall, T. J., & Barker, P. M. (2020). A pressure-invariant Neutral Density variable for the World's Oceans. *Journal of Physical Oceanography*, 50(12), 3585-3604. <https://doi.org/10.1175/JPO-D-19-0321.1>
- **Stanley, G. J.**, Dowling, T. E., Bradley, M. E., & Marshall, D. P. (2020). Ertel Potential Vorticity versus Bernoulli Potential on Approximately Neutral Surfaces in the Antarctic Circumpolar Current. *Journal of Physical Oceanography*, 50(9), 2621-2648. <https://doi.org/10.1175/JPO-D-19-0140.1>
- Groeskamp, S., de Lavergne, C., Holmes, R., Tamsitt, V., Frenger, I., Chapman, C. C., Newsom, E., & **Stanley, G. J.** (2019). Climate Recorded in Seawater: A Workshop on Water-Mass Transformation Analysis for Ocean and Climate Studies. *Bulletin of the American Meteorological Society*, 100(9), ES243-ES247. <https://doi.org/10.1175/BAMS-D-19-0153.1>
- Padget, O., **Stanley, G.**, Willis, J. K., Fayet, A. L., Bond, S., Maurice, L., Shoji, A., Dean, B., Kirk, H., Juarez-Martinez, I., Freeman, R., Bolton, M., & Guilford, T. (2019). Shearwaters know the direction and distance home but fail to encode intervening obstacles after free-ranging foraging trips. *Proceedings of the National Academy of Sciences*, 116(43), 21629-21633. <https://doi.org/10.1073/pnas.1903829116>
- **Stanley, G. J.** (2019b). The exact geostrophic streamfunction for neutral surfaces. *Ocean Modelling*, 138, 107-121. <https://doi.org/10.1016/j.ocemod.2019.04.002>
- **Stanley, G. J.** (2019a). Neutral surface topology. *Ocean Modelling*, 138, 88-106. <https://doi.org/10.1016/j.ocemod.2019.01.008>
- Cheng, R., Jackson, D. M., & **Stanley, G. J.** (2018). Combinatorial Aspects of the Quantized Universal Enveloping Algebra of $\mathfrak{sl}_{n+1}$. *Annals of Combinatorics*, 22(4), 681-710. <https://doi.org/10.1007/s00026-018-0404-2>
- **Stanley, G. J.**, & Saenko, O. A. (2014). Bottom-Enhanced Diapycnal Mixing Driven by Mesoscale Eddies: Sensitivity to Wind Energy Supply. *Journal of Physical Oceanography*, 44(1), 68-85. <https://doi.org/10.1175/JPO-D-13-0116.1>

PREPRINTS / NON-PEER REVIEWED ARTICLES

- Nurser, A. J. G., Griffies, S. M., Zika, J. D., & **Stanley, G. J.** (2022). A mathematical formalism for circulation in water mass configuration space. *ESS Open Archive*. <https://doi.org/10.1002/essoar.10511370.2>
- **Stanley, G.**, 2014: The most minimal seed for transition to turbulence in shear flow. Tech. Rep. *Proceedings Volume 2014*, Woods Hole Oceanographic Institution, 359-384 pp.

CONFERENCE PROCEEDINGS / SEMINARS / WORKSHOPS

CONFERENCE PRESENTATIONS: ORAL

- **Stanley G. J.**, T. J. McDougall, and P. M. Barker, Algorithmic improvements to finding approximately neutral surfaces Ocean Sciences Meeting, Abstract Number 3737, 28 February-4 March 2022.

- **Stanley G. J.**, The topology of neutral surfaces and their exact geostrophic streamfunction Ocean Sciences Meeting, Abstract Number PL53B-06, 16–21 February 2020. San Diego, CA, USA.
- **Stanley G. J.**, The topology of neutral surfaces and their exact geostrophic streamfunction, IUGG General Assembly, 8–18 July, 2019. Montreal, Canada.
- **Stanley G. J.**, Neutral Surface Topology, Physical Oceanography Dissertation Symposium X, 21–25 October 2018. Kona, HI, USA.
- **Stanley G. J.**, An exact geostrophic stream function on a neutral surface, Ocean Modelling Group, 9 September 2016. Liverpool, UK.
- **Stanley G. J.** and D. P. Marshall, Inferring Large-Scale Bottom Velocity from Sparse Data, Ocean Modelling Group, 21–22 September 2015. Corpus Christi College, Cambridge, UK.
- **Stanley G. J.** and D. P. Marshall, Predicting Bottom Velocities from Deep ARGO, IUGG General Assembly, 22 June–2 July 2015. Prague, Czech Republic.
- **Stanley G. J.** and O. A. Saenko, Diapycnal mixing parameterized by energy release from mesoscale eddies, IAHS-IAPSO-IASPEI Joint Assembly, Abstract Number Po3S3.06, 22–26 July 2013. Gothenburg, Sweden.

CONFERENCE PRESENTATIONS: POSTERS

- **Stanley, G. J.**, T. E. Dowling, M. E. Bradley, D. P. Marshall, Ertel Potential Vorticity versus Bernoulli Potential on Approximately Neutral Surfaces in the Antarctic Circumpolar Current, IUGG General Assembly, 8–18 July, 2019. Montreal, Canada.
- **Stanley G. J.** and D. P. Marshall, Inferring Large-Scale Bottom Velocity from Sparse Data, Ocean Sciences Meeting, Abstract Number OD14B-2421, 21–26 February 2016. New Orleans, LA, USA.
- **Stanley G. J.** and O. A. Saenko, On the Energetics of Oceanic Mesoscale Eddies and their Parameterization Modified to Induce Diapycnal Mixing, AGU Fall Meeting, 3–7 December 2012. San Francisco, CA, USA.
- **Stanley G. J.** and O. A. Saenko, On the Energetics of Oceanic Mesoscale Eddies and their Parameterization Modified to Induce Diapycnal Mixing, Graduate Climate Conference, 26–28 October 2012. Pack Forest, WA, USA.

SEMINARS

- **Stanley G. J.**, Untangling multi-valued functional relations using the Reeb graph on oceanic neutral surfaces, June 2, 2023. Aspen Center for Physics, Aspen CO, USA.
- **Stanley G. J.**, Using Topology to learn about Neutral Surfaces in the Ocean, January 31, 2023. University of Victoria, Victoria BC, Canada.
- **Stanley G. J.**, Topology and Optimization for Neutral Surfaces in the Ocean, *One World Mathematics of Climate*, April 5, 2022. Online global seminar series.
- **Stanley G. J.**, The topology of neutral surfaces and their exact geostrophic streamfunction, September 3, 2019. University of Toronto, Toronto ON, Canada.
- **Stanley G. J.**, The topology of neutral surfaces and their exact geostrophic streamfunction, August 2, 2019. Woods Hole Oceanographic Institution Geophysical Fluid Dynamics Summer School, Woods Hole, MA, USA.
- **Stanley G. J.**, The topology of neutral surfaces and their exact geostrophic streamfunction, May 21, 2019. Institute for Marine and Antarctic Studies, Hobart, Australia.
- **Stanley G. J.**, The topology of neutral surfaces and their exact geostrophic streamfunction, 7 March 2019. University of New South Wales, Sydney, Australia.
- **Stanley G. J.** and N. R. Lebovitz, The Most Minimal Seed for the Onset of Shear Turbulence, Geophysical Fluid Dynamics Program, Woods Hole Oceanographic Institution, 19 August 2014. Woods Hole, MA, USA.
- **Stanley G. J.** and O. A. Saenko, From Winds to Eddies to Diapycnal Mixing over Topography: Driving the Abyssal Meridional Overturning Circulation, National Oceanography Centre, 19 March 2014. Southampton, UK.

WORKSHOPS

- Ocean Model Development Panel (OMDP) Community for the Numerical Modeling of the Global, Regional, and Coastal Ocean (COMMODORE) Workshop, 9–12 September 2024. NCAR Mesa Laboratory, Boulder CO, USA.
- Geometric and Field Theoretic Methods for Astro-, Geo-, and Bio-physical Fluids, 29 May – 9 June 2023. Aspen Center for Physics, Aspen CO, USA.
- Multiscale Dynamics of the Southern Ocean, 4–5 July 2022. Australian National University, Canberra, Australia.
- Water Mass Transformation for Ocean Physics and Biogeochemistry, 4–6 February 2019. UNSW Sydney, NSW, Australia.

SOFTWARE CONTRIBUTIONS

- `neutral-surfaces`: MATLAB software for computing topobaric surfaces, omega surfaces, geostrophic streamfunctions. Available at <https://github.com/geoffstanley/neutral-surfaces>
- `neutralocean`: Python software for computing omega surfaces. Available at <https://github.com/geoffstanley/neutralocean>

GRANTS

- Banting Postdoctoral Fellowships Program 2021-09-22
140,000 CAD
Oct 2022 - Sept 2024
Understanding geophysical turbulence and planetary shockwaves for prediction of Sudden Stratospheric Warming events

ACADEMIC MENTORING EXPERIENCE

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|-----------------------|--------------------------------------|--|
| • Jan 2019 – Feb 2023 | Aaron Lang (co-advisor) | UNSW Mathematics PhD student |
| • Feb – Nov 2021 | Josef Bisits (co-advisor) | UNSW Mathematics Honours student (First Class) |
| • Feb – Nov 2020 | Alexander Robinson (primary advisor) | UNSW Mathematics Honours student (First Class) |
| • May – Oct 2020 | Garrett Finucane (primary advisor) | University of Washington, summer research assistant |
| • June – Aug 2019 | Houssam Yassin (co-advisor) | Woods Hole Oceanographic Institution,
Geophysical Fluid Dynamics summer school Fellow |

TEACHING EXPERIENCE

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| • 2023 | Lecturer and coordinator of multivariate calculus and ordinary differential equations (90 students, 2 TAs) | MATH 202, U Victoria |
| • 2016 | Tutor for Geophysical Fluid Dynamics | Masters course in Mathematical Physics, U Oxford |
| • 2014 | Tutor for Flows, Fluctuations, and Complexity | 3rd year course B1, Mansfield College, U Oxford |
| • 2012 | Lab Instructor for Oceans and Atmospheres | EOS 110, U Victoria |
| • 2011 | Marker for Earth System Modelling | EOS 225, U Victoria |
| • 2007 | Marker for Advanced Calculus 1 | MATH 147, U Waterloo |

SELECTED ACADEMIC AWARDS

- 2018 Physical Oceanography Dissertation Symposium (PODS) Invited Participant National Science Foundation (US)
- 2014 Geophysical Fluid Dynamics Fellowship Woods Hole Oceanographic Institution
- 2013 — 2016 Clarendon Fund Scholarship U Oxford
- 2013 — 2016 Canadian Alumni Scholarship Linacre College, Oxford
- 2012 Gagnon Memorial Scholarship U Victoria
- 2011 NSERC Julie Payette Research Scholarship Canada
(to the top 24 applicants for MSc funding)
- 2011 K.D. Fryer Gold Medal U Waterloo
(to one of the top Math graduates also exemplifying good student citizenship)
- 2006 — 2010 René Descartes Scholarship U Waterloo
- 2006 — 2010 Queen Elizabeth II Aiming for the Top Scholarship Ontario
- 2007, 2008, 2010 NSERC Undergraduate Student Research Awards Canada
- 2009 Robert Schaefer Memorial Award U Waterloo
- 2008 President's Research Award U Waterloo

PROFESSIONAL ACTIVITIES

- 2019 Staff member at the Geophysical Fluid Dynamics summer school, Woods Hole Oceanographic Institution
- 2015 Scientist aboard the *RRS Discovery* on the Extended Ellett Line
- 2015 Organizer of a 2-day Software Carpentry workshop, AOPP, U Oxford
- 2013 — 2015 Graduate Student Representative, U Oxford
- 2010 Undergraduate Representative to the Chair Selection Committee
Department of Combinatorics and Optimization, University of Waterloo
- 2010 Conference Staff Volunteer, Canadian Undergraduate Mathematics Conference
- 2015 — ongoing Peer reviewer for the Journal of Physical Oceanography, Fluids, Ocean Sciences, Journal of Geophysical Research: Oceans, Journal of Marine Science and Engineering, Journal of Climate, and the National Science Foundation (USA).

RESEARCH ASSISTANTSHIPS

- 2010 Department of Combinatorics and Optimization University of Waterloo
Project: *Straightening formulas in the quantized universal enveloping algebra sl_2*
Supervisor: David M. Jackson
- 2008 Department of Physics and Astronomy, University of Waterloo
Project: *Numerical simulation of entropy evolution in merging galaxy clusters*
Supervisor: Michael Balogh
- 2007 Institute for Quantum Computing, University of Waterloo
Project: *Numerical simulation of a superconducting flux qubit and the fidelity of its quantum state*
Supervisor: Frank Wilhelm
- 2006 Richard Lewar Centre for Excellence, Heart & Stroke Lab, University of Toronto
Project: *Examining the electrophysiological structure of cardiac sodium ion channels*
Supervisor: Peter Backx

EXTRACURRICULAR EXPERIENCE

- 2014 — 2015 Green Student: Technical & Financial Support
Linacre College, Oxford
- 2012 — 2013 Co-Coordinator of Café Scientifique, a grassroots public science series
Faculty of Science, University of Victoria
- 2008 — 2010 Residence Don, University of Waterloo Housing and Residence
Leader and role model to over 100 students for 3 terms; awarded "Rookie of the Term"